

AI Roadmap

A time-phased plan linking AI initiatives to capability, dependency and business outcome

GLOSSARY 04 - SHEET 2 / 8

PROFESSIONAL DEFINITION

An AI roadmap is a time-phased plan that links a sequence of AI initiatives to the capabilities each requires, the dependencies between them, and the business outcomes each is intended to produce. It is a coordination instrument rather than a schedule: its function is to make visible which capability must exist before which initiative can proceed, so that sequencing errors are detected in planning rather than in delivery (Phaal, Farrukh and Probert, 2004).

WHO DEFINED IT ORIGINALLY

Technology roadmapping was developed at Motorola under Robert Galvin in the late 1970s and formalised as the T-Plan method by Robert Phaal, Clare Farrukh and David Probert at the University of Cambridge Institute for Manufacturing, published in Technological Forecasting and Social Change in 2004.

WHY THIS DEFINITION WAS DEVELOPED

Technology plans and business plans were produced by different functions on different cycles and were reconciled only when a delivery date was missed. Roadmapping placed market, product and technology layers on a single time axis so that the dependency between them could be inspected before commitment.

FIGURE 1 - THREE LAYERS ON ONE TIME AXIS



Figure 1. The roadmap is read vertically as well as horizontally. An initiative whose enabling capability arrives after its own start date is a scheduling error visible in planning; discovered in delivery it is an overrun (after Phaal, Farrukh and Probert, 2004).

HOW IT IS USED

Lay out three layers - business outcome, initiative, enabling capability - on one horizontal time axis, and read upward from capability to outcome to test whether each initiative's prerequisites will exist when it starts.

WHEN IT IS USED

At the start of a multi-year AI programme, at annual replanning, when a delivery date slips for reasons of missing data or platform rather than model work, and when a new initiative is proposed mid-cycle.

BY WHOM IT IS USED

Chief data and AI officers, portfolio and programme offices, platform and data engineering leaders supplying the capability layer, and business sponsors owning the outcome layer.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor artefact was the project Gantt chart, which sequences tasks within one initiative and is silent about dependencies across initiatives or about the capability each presupposes. The multi-layer roadmap makes the cross-initiative dependency explicit and forces the capability layer to be stated, which is precisely where AI programmes fail: the model is ready and the governed data feed is not.

RELATED DEFINITIONS IN THIS COURSE

[AI Strategy - Unit 3](#)
[AI Portfolio Management - Unit 3](#)
[MLOps - Unit 4](#)
[Benefits Realisation - Unit 4](#)
[Digital Maturity - Unit 1](#)

SOURCES (HARVARD REFERENCING STYLE)

Phaal, R., Farrukh, C.J.P. and Probert, D.R. (2004) 'Technology roadmapping: a planning framework for evolution and revolution', Technological Forecasting and Social Change, 71(1-2), pp. 5-26.

Phaal, R., Farrukh, C.J.P. and Probert, D.R. (2001) T-Plan: The Fast Start to Technology Roadmapping. Cambridge: University of Cambridge Institute for Manufacturing.

Willyard, C.H. and McClees, C.W. (1987) 'Motorola's technology roadmap process', Research Management, 30(5), pp. 13-19.